

# Lab\_Syn\_3.1

You have to synthesize **ALU\_Top hierarchy** files using **Design Compiler shell** and generate **Technology Dependent** gate level netlists in Verilog format using standard cell libraries (typical, worst, best) inside **std\_cells** folder:-

- **scmetro\_tsmc\_cl013g\_rvt\_tt\_1p2v\_25c.db**
- **scmetro\_tsmc\_cl013g\_rvt\_ff\_1p32v\_m40c**
- **scmetro\_tsmc\_cl013g\_rvt\_ss\_1p08v\_125c**

## Steps: -

1. Copy **Lab\_Syn\_3.1** folder from your computer into the virtual machine inside **Labs** directory
2. Open **syn\_script.tcl** and **Update variable "flattened" to 0** and run **syn\_script.tcl** script through design compiler shell using the following command in the terminal `>> dc_shell -f syn_script.tcl | tee syn.log`
  - Open the GUI and view the schematic
3. Open **syn\_script.tcl** and **Update variable "flattened" to 1** and run **syn\_script.tcl** script through design compiler shell using the following command in the terminal `>> dc_shell -f syn_script.tcl | tee syn.log`
  - Open the GUI and view the schematic
4. What is the difference between the 2 netlists **ALU\_netlist\_flattened.v** & **ALU\_netlist\_hierarchy.v** and which command make this difference ?
5. View the 2 Area reports **area\_flattened.v** Verses **area\_hierarchy.v**